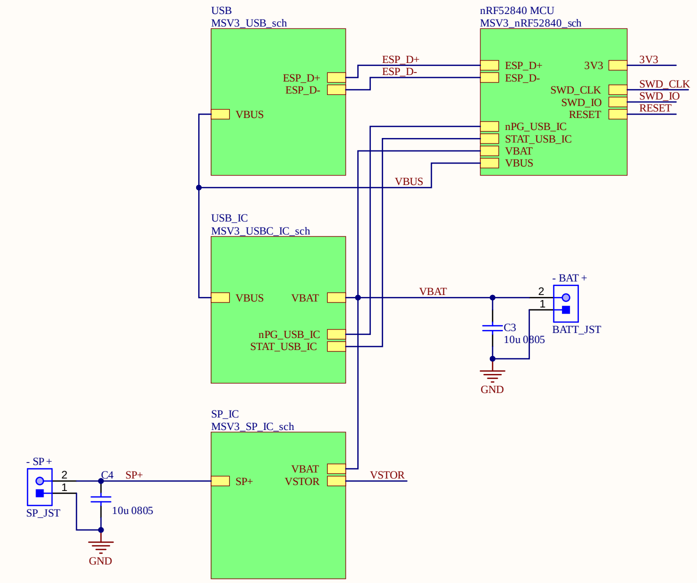
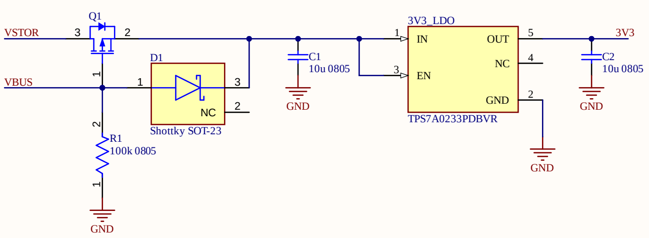
REVISION	DESCRIPTION	DATE	APPROVED

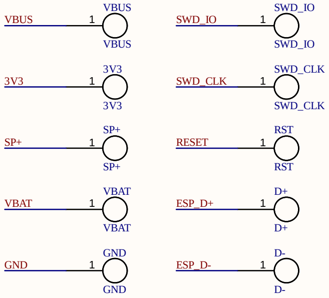
TOP LEVEL


USB
MSV3_USB_sch
ESP_D+
ESP_D-
VBUS
nRF52840 MCU
MSV3_nRF52840_sch
ESP_D+
ESP_D-
3V3
SWD_CLK
SWD_IO
RESET
nPG_USB_IC
STAT_USB_IC
VBAT
VBUS
3V3
SWD_CLK
SWD_IO
RESET
VBUS
USB_IC
MSV3_USBC_IC_sch
VBUS
VBAT
nPG_USB_IC
STAT_USB_IC
VBAT
- BAT +
2
1
BATT_JST
C3
10u 0805
GND
SP_IC
MSV3_SP_IC_sch
VBAT
VSTOR
SP+
VSTOR
- SP +
2
1
SP_JST
C4
SP+
10u 0805
GND

Power Mux


PMOS SOT-23
Q1
VSTOR
3
2
1
VBUS
D1
1
3
2
NC
Shottky SOT-23
2
R1
100k 0805
1
GND
C1
10u 0805
GND
3V3_LDO
1
IN
OUT
5
3
EN
NC
4
GND
2
TPS7A0233PDBVR
GND
3V3
C2
10u 0805
GND

TPs

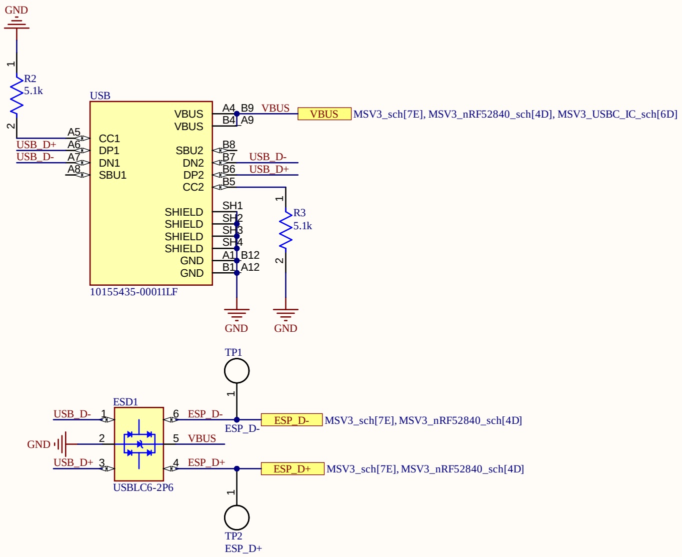

VBUS
1
VBUS
VBUS
SWD_IO
1
SWD_IO
SWD_IO
3V3
1
3V3
3V3
SWD_CLK
1
SWD_CLK
SWD_CLK
SP+
1
SP+
SP+
RESET
1
RST
RST
VBAT
1
VBAT
VBAT
ESP_D+
1
D+
D+
GND
1
GND
GND
ESP_D-
1
D-
D-

APPROVALS | DATE
ENG: *
DSN: *
CHK: *
REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

PROJECT
Altium™
*
*
*
*
PROJECT REVISION: DOCUMENT REVISION: DESIGN ITEM: 4349b493585d4387caf0[illegible]cca2786d78c04afd
TITLE
*
SIZE B
CAGE CODE
DWG NO.
REV
SCALE:
FILE NAME MSV3_sch.SchDoc
SHEET 1 OF 5

THIS DOCUMENT AND THE DATA DISCLOSED HEREIN OR HEREWITH IS THE PROPERTY OF ALTIUM LIMITED AND MAY BE FREELY DISTRIBUTED IN WHOLE. NO RIGHTS ARE RESERVED OR EXPRESS OR IMPLIED WARANTEE GIVEN.

REVISION	DESCRIPTION	DATE	APPROVED


GND
1
R2
5.1k
2
USB
A5
USB_D+
A6
USB_D-
A7
A8
CC1
DP1
DN1
SBU1
VBUS
VBUS
SBU2
DN2
DP2
CC2
SHIELD
SHIELD
SHIELD
SHIELD
GND
GND
A4 B9
B4 A9
VBUS
VBUS
MSV3_sch[7E], MSV3_nRF52840_sch[4D], MSV3_USBC_IC_sch[6D]
B8
B7
USB_D-
B6
USB_D+
B5
SH1
SH2
SH3
SH4
A1 B12
B1 A12
1
R3
5.1k
2
10155435-00011LF
GND
GND
TP1
1
ESD1
USB_D-
1
6
ESP_D-
ESP_D-
ESP_D-
MSV3_sch[7E], MSV3_nRF52840_sch[4D]
GND
2
5
VBUS
USB_D+
3
4
ESP_D+
ESP_D+
MSV3_sch[7E], MSV3_nRF52840_sch[4D]
USBLC6-2P6
1
TP2
ESP_D+

APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	

REFERENCE DOCUMENTS

BOM:

ASSY DWG:

FAB DWG:

PCB DWG:

PROJECT

*
*
*
*

PROJECT REVISION: 4349b493585d4387caf0 DOCUMENT REVISION: DESIGN ITEM: cca2786d78c04afdc

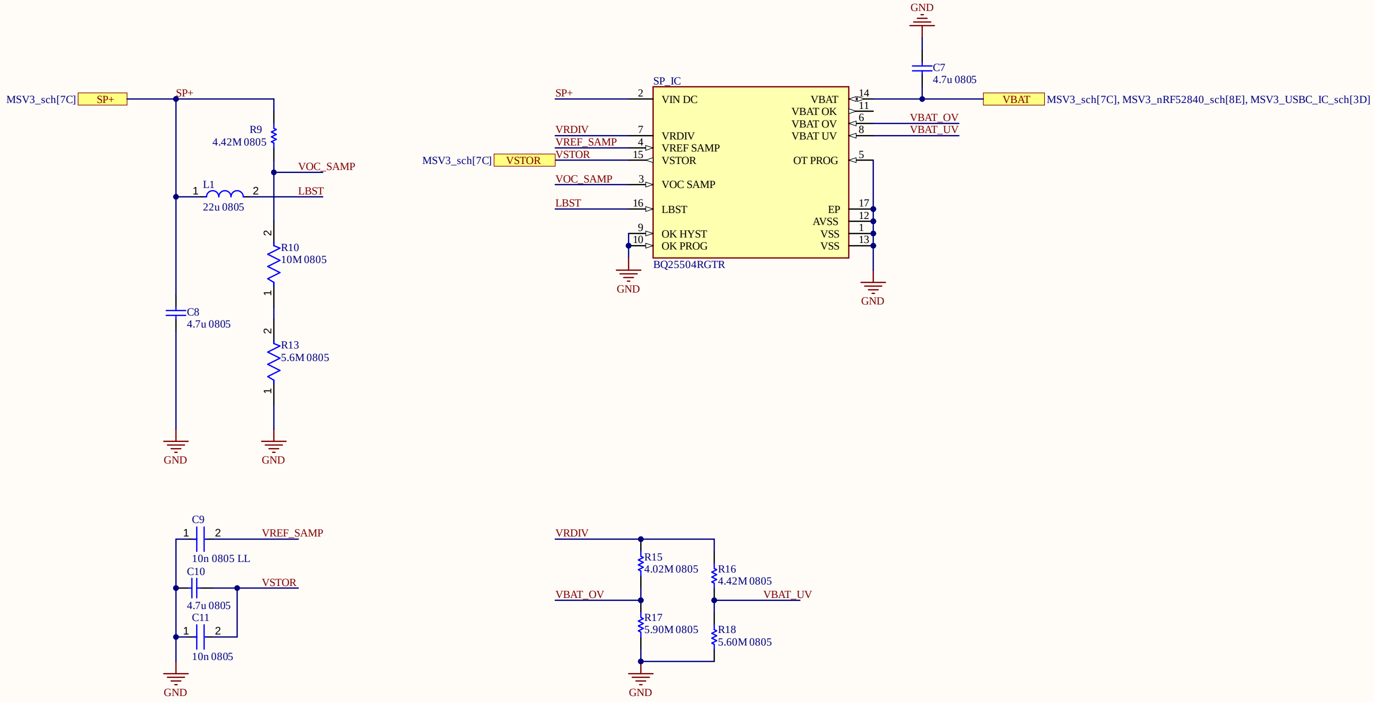
TITLE

*

SIZE	CAGE CODE	DWG NO.	REV
B			

SCALE: FILE NAME MSV3_USB_sch.SchDoc SHEET 2 OF 5

REVISION	DESCRIPTION	DATE	APPROVED


MSV3_sch[7C]
SP+
SP+
R9
4.42M 0805
VOC_SAMP
L1
1
2
22u 0805
LBST
2
R10
10M 0805
1
C8
4.7u 0805
2
R13
5.6M 0805
1
GND
GND
SP_IC
SP+
2
VIN DC
VRDIV
7
VRDIV
VREF_SAMP
4
VREF SAMP
MSV3_sch[7C]
VSTOR
VSTOR
15
VSTOR
VOC_SAMP
3
VOC SAMP
LBST
16
LBST
9
OK HYST
10
OK PROG
GND
BQ25504RGTR
VBAT
14
VBAT OK
11
VBAT OV
6
VBAT_OV
VBAT UV
8
VBAT_UV
OT PROG
5
EP
17
AVSS
12
VSS
1
VSS
13
GND
GND
C7
4.7u 0805
VBAT
MSV3_sch[7C], MSV3_nRF52840_sch[8E], MSV3_USBC_IC_sch[3D]
C9
1
2
VREF_SAMP
10n 0805 LL
C10
VSTOR
4.7u 0805
C11
1
2
10n 0805
GND
VRDIV
R15
4.02M 0805
R16
4.42M 0805
VBAT_OV
VBAT_UV
R17
5.90M 0805
R18
5.60M 0805
GND

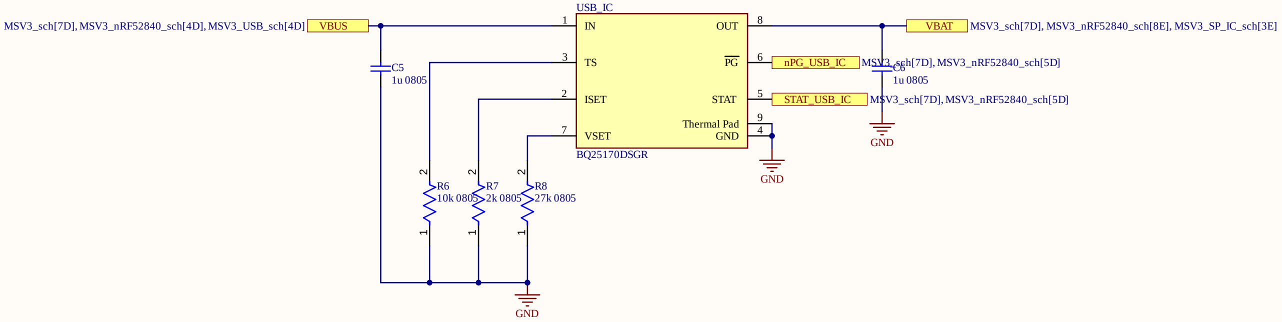
APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	

REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

PROJECT * * * *
PROJECT REVISION: 4349b493585d4387caf0 [illegible] cca2786d78c04afdc
DOCUMENT REVISION:
DESIGN ITEM:
TITLE *
SIZE B
CAGE CODE
DWG NO.
REV
SCALE:
FILE NAME MSV3_SP_IC_sch.SchDoc
SHEET 3 OF 5

DWG. NO. REV. SHT. 3

REVISION	DESCRIPTION	DATE	APPROVED


MSV3_sch[7D], MSV3_nRF52840_sch[4D], MSV3_USB_sch[4D]
VBUS
C5
1u 0805
USB_IC
1
IN
3
TS
2
ISET
7
VSET
BQ25170DSGR
OUT
8
PG
6
STAT
5
Thermal Pad
9
GND
4
GND
VBAT
MSV3_sch[7D], MSV3_nRF52840_sch[8E], MSV3_SP_IC_sch[3E]
nPG_USB_IC
MSV3_sch[7D], MSV3_nRF52840_sch[5D]
STAT_USB_IC
MSV3_sch[7D], MSV3_nRF52840_sch[5D]
C6
1u 0805
GND
R6
10k 0805
R7
2k 0805
R8
27k 0805
2
1
GND

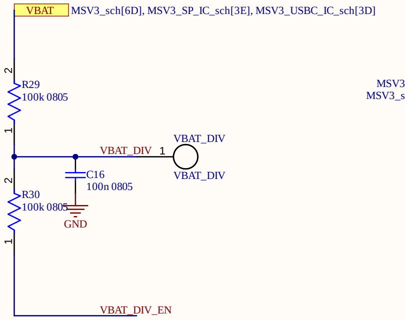
APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	

REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

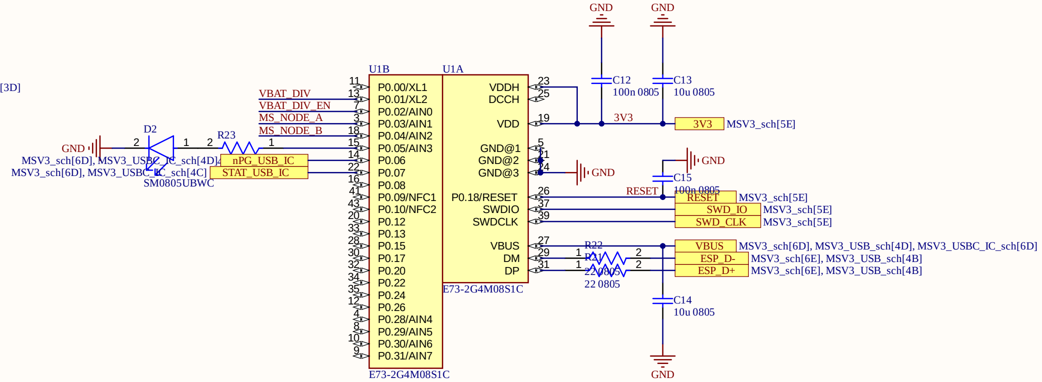
PROJECT
Altium
*
*
*
*
PROJECT REVISION: 4349b493585d4387caf0[illegible]cca2786d78c04afd
DOCUMENT REVISION:
DESIGN ITEM:
TITLE
*
SIZE B
CAGE CODE
DWG NO.
REV
SCALE:
FILE NAME MSV3_USBC_IC_sch.SchDoc
SHEET 4 OF 5

REVISION	DESCRIPTION	DATE	APPROVED

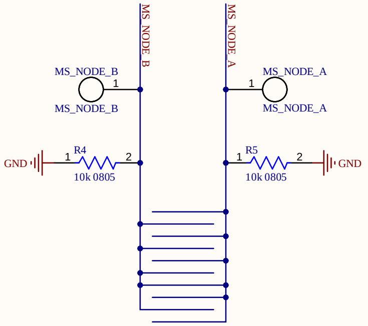
BATTERY LEVEL


VBAT
MSV3_sch[6D], MSV3_SP_IC_sch[3E], MSV3_USBC_IC_sch[3D]
R29
100k 0805
2
1
VBAT_DIV
1
VBAT_DIV
VBAT_DIV
C16
100n 0805
GND
R30
100k 0805
2
1
VBAT_DIV_EN

nRF52840 MCU


U1B
U1A
VBAT_DIV
VBAT_DIV_EN
MS_NODE_A
MS_NODE_B
GND
D2
SM0805UBWC
2
1
R23
2
1
MSV3_sch[6D], MSV3_USBC_IC_sch[4D]
nPG_USB_IC
MSV3_sch[6D], MSV3_USBC_IC_sch[4C]
STAT_USB_IC
11 P0.00/XL1
13 P0.01/XL2
7 P0.02/AIN0
3 P0.03/AIN1
18 P0.04/AIN2
15 P0.05/AIN3
14 P0.06
22 P0.07
16 P0.08
41 P0.09/NFC1
43 P0.10/NFC2
20 P0.12
33 P0.13
28 P0.15
30 P0.17
32 P0.20
34 P0.22
35 P0.24
12 P0.26
4 P0.28/AIN4
8 P0.29/AIN5
10 P0.30/AIN6
9 P0.31/AIN7
E73-2G4M08S1C
VDDH 23
DCCH 25
VDD 19
GND@1 5
GND@2 21
GND@3 24
P0.18/RESET 26
SWDIO 37
SWDCLK 39
VBUS 27
DM 29
DP 31
E73-2G4M08S1C
GND
GND
C12
100n 0805
C13
10u 0805
3V3
3V3
MSV3_sch[5E]
GND
GND
C15
100n 0805
RESET
RESET
MSV3_sch[5E]
SWD_IO
MSV3_sch[5E]
SWD_CLK
MSV3_sch[5E]
R22
R21
22 0805
22 0805
1
2
1
2
VBUS
MSV3_sch[6D], MSV3_USB_sch[4D], MSV3_USBC_IC_sch[6D]
ESP_D-
MSV3_sch[6E], MSV3_USB_sch[4B]
ESP_D+
MSV3_sch[6E], MSV3_USB_sch[4B]
C14
10u 0805
GND

MS_NODE


MS_NODE_B
MS_NODE_A
MS_NODE_B
1
MS_NODE_B
MS_NODE_A
1
MS_NODE_A
GND
R4
1
2
10k 0805
R5
1
2
10k 0805
GND

APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	

REFERENCE DOCUMENTS

BOM:

ASSY DWG:

FAB DWG:

PCB DWG:

PROJECT

*
*
*
*

PROJECT REVISION: 4349b493585d4387caf02a02c13652d631

DOCUMENT REVISION:

DESIGN ITEM:

TITLE

*

SIZE	CAGE CODE	DWG NO.	REV
B			

SCALE:

FILE NAME MSV3_nRF52840_sch.SchDoc

SHEET 5 OF 5

DWG. NO. REV. 5 SHT.